

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0615	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	K. Bhanu Prakash	Reg. No.:	192525184
Section / Batch:	2025	Date:	17-7-26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Apply Master Theorem and asymptotic – Q2
analysis to real-world scenarios Q1 –50 Q2 –50

Application-Based Questions

SECTION A – APPLICATION BASED QUESTIONS

Q 1. Robotics Based on AI Expert Systems (Detailed Explanation)

Robotics is a branch of engineering and Artificial Intelligence (AI) that deals with the design, construction, operation, and application of robots. A **robot** is an intelligent machine that can perform tasks automatically or with minimal human intervention. Modern robots use AI techniques to understand their environment, make decisions, and complete tasks efficiently.

Program Code:



```
1 import java.util.Scanner;
2
3 public class RobotExpertSystem {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.println("=== ROBOTICS AI EXPERT SYSTEM ===");
10
11         System.out.print("Is an obstacle detected? (yes/no): ");
12         String obstacle = sc.nextLine();
13
14         System.out.print("Is the battery low? (yes/no): ");
15         String battery = sc.nextLine();
16
17         System.out.print("Is a human detected? (yes/no): ");
18         String human = sc.nextLine();
19
20         System.out.print("Is the path clear? (yes/no): ");
21         String path = sc.nextLine();
22
23         System.out.println("\n=== EXPERT SYSTEM DECISION ===");
24
25         // Rule 1
26         if (battery.equalsIgnoreCase("yes")) {
27             System.out.println("Decision: Return to charging station.");
28         }
29
30         // Rule 2
31
32         // Rule 2
33         else if (human.equalsIgnoreCase("yes")) {
34             System.out.println("Decision: Stop and interact with the human.");
35         }
36
37         // Rule 3
38         else if (obstacle.equalsIgnoreCase("yes")) {
39             System.out.println("Decision: Obstacle detected.");
40             System.out.println("Action: Turn left or right to avoid obstacle.");
41         }
42
43         // Rule 4
44         else if (path.equalsIgnoreCase("yes")) {
45             System.out.println("Decision: Path is clear.");
46             System.out.println("Action: Move forward.");
47         }
48
49         // Default Rule
50         else {
51             System.out.println("Decision: Stop and analyze the environment.");
52         }
53
54         sc.close();
55     }
56 }
```

Output

[Clear](#)

```
=== ROBOTICS AI EXPERT SYSTEM ===
Is an obstacle detected? (yes/no): yes
Is the battery low? (yes/no): no
Is a human detected? (yes/no): yes
Is the path clear? (yes/no): yes

=== EXPERT SYSTEM DECISION ===
Decision: Obstacle detected.
Action: Turn left or right to avoid obstacle.

=== Code Execution Successful ===
```

Q2. Application: Online Learning Recommendation System

An online learning platform processes user interaction data using the recurrence $T(n) = T(n-1) + n$. Apply the substitution method to expand the recurrence step-by-step and derive the final time complexity. Clearly show intermediate steps and simplify the expression. Analyze how the algorithm performs when the number of users grows rapidly. Interpret the efficiency using asymptotic notation and discuss whether the system is scalable for large educational platforms.

Program Code:

RoboticsAIExpertSystem.java

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```
1- import java.util.Scanner;
2
3- public class RoboticsAIExpertSystem {
4
5-     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.println("===== ROBOTICS WITH AI EXPERT SYSTEM =====");
10        System.out.println("1. Healthcare Robot");
11        System.out.println("2. Industrial Robot");
12        System.out.println("3. Agriculture Robot");
13        System.out.println("4. Home Service Robot");
14        System.out.println("5. Security Robot");
15
16        System.out.print("\nEnter application number: ");
17        int choice = sc.nextInt();
18
19        System.out.println("\n===== EXPERT SYSTEM DECISION =====");
20
21        if (choice == 1) {
22            System.out.println("Application: Healthcare Robot");
23            System.out.println("AI Decision: Monitor patient health.");
24            System.out.println("Action: Alert doctor if abnormal condition is detected.");
25        }
26
27        else if (choice == 2) {
28            System.out.println("Application: Industrial Robot");
29            System.out.println("AI Decision: Check machine condition.");
30        }
31    }
32 }
```

```

29         System.out.println("AI Decision: Check machine condition.");
30         System.out.println("Action: Stop machine if a dangerous fault is
           detected.");
31     }
32
33     else if (choice == 3) {
34         System.out.println("Application: Agriculture Robot");
35         System.out.println("AI Decision: Check soil and crop condition.");
36         System.out.println("Action: Water the crops if the soil is dry.");
37     }
38
39     else if (choice == 4) {
40         System.out.println("Application: Home Service Robot");
41         System.out.println("AI Decision: Check the environment.");
42         System.out.println("Action: Clean the room and avoid obstacles.");
43     }
44
45     else if (choice == 5) {
46         System.out.println("Application: Security Robot");
47         System.out.println("AI Decision: Detect suspicious activity.");
48         System.out.println("Action: Send an alert to the security team.");
49     }
50
51     else {
52         System.out.println("Invalid application choice.");
53     }
54
55     sc.close();
56 }
57 }

```

Output:

Output

Clear

===== ROBOTICS WITH AI EXPERT SYSTEM =====

1. Healthcare Robot
2. Industrial Robot
3. Agriculture Robot
4. Home Service Robot
5. Security Robot

Enter application number: 2

===== EXPERT SYSTEM DECISION =====

Application: Industrial Robot

AI Decision: Check machine condition.

Action: Stop machine if a dangerous fault is detected.

=== Code Execution Successful ===

Code of Conduct

I K. Bhanu Prakash (192525184) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Clearly	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	identifies all requirements	Good	understanding with minor gaps	or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Proficient	Basic	Poor understanding
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach			
Accuracy of Solution	20%	Completely correct solution with proper justification			
Clarity	10%	Clear, well-organized, and properly explained solution			

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Dr G Nagarajan		
Signature:	Signature:	Signature:
Date:	Date:	Date: